MIAX Options Exchange

Administrative Information Subscriber Feed

AIS Interface Specification

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Version 2.6b

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Table of Contents

1. Overview 3

1.1 Exchange related information 3
1.1.1 Hours of operation for MIAX Options Exchange 3
1.1.2 Obtaining more information 4

1.2 Testing of AIS Subscription 4

1.3 Answers to FAQs 4

1.4 Data Types 5

2. AIS Architecture 6

3. Session Level Protocol 8

3.1 Real-time AIS Feed 8

3.2 AIS Retransmission Interface 8
3.2.1 SesM Gap Fill Service 8
3.2.2 Last Value Refresh Service 8
3.2.3 Session Termination 9

4. Application Message Formats 10

4.1 MIAX System Time Message 10

4.2 Simple Series Update 10

4.3 Complex Strategy Definition Update 13

4.4 System State 14

4.5 Simple Liquidity Seeking Event Notification 15

4.6 Complex Liquidity Seeking Event Notification 17

4.7 Underlying Trading Status Notification 19

4.8 Theoretical Settlement Reference Price Notification 19

4.9 Theoretical Settlement Price Notification 21

Appendix A: MIAX AIS Subscription/Connectivity Information 23

Appendix B: Contact List 24

Appendix C: Liquidity Seeking Event Notification Details 25

Appendix D: Revision History 31

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1. Overview

MIAX Administrative Information Subscriber Feed (**AIS**) is a data feed that allows subscribers to receive real-time updates of the following information from the MIAX Options Market

- Products traded on MIAX
- Trading status for MIAX and products traded on MIAX
- Liquidity seeking event notifications

AIS Features

AIS messaging and the system architecture are designed for low latency messaging. Some of the key features of the interface are:

- AIS is offered with redundant multicast feeds (A Feed & B Feed) to provide single point of failure hardware and network fault tolerance and to provide an opportunity for recipients to arbitrate the two feeds to auto-fill gaps.
- AIS real-time messages are disseminated over multicast to achieve a fair delivery mechanism. AIS requires the use of MIAX proprietary SesM over TCP/IP protocol for retransmission lines in order to provide a **guaranteed delivery** mechanism for gap fills.
- The AIS retransmission service also provides a 3.2.2 Last Value Refresh Service to facilitate fast intra-day recovery without a full day gap fill.
- AIS notifications provide current **electronic system status** allowing the subscribers to take necessary actions immediately.

This specification is intended to be used by MIAX AIS Feed subscribers only.

1.1 Exchange related information

1.1.1 Hours of operation for MIAX Options Exchange

Please refer to the MIAX website for details about times for each of these events.

Note: Times specified below are in United States Eastern Time zone.

Start of Session: Start of dissemination of messages. After 5:00 a.m.

Trading Session for Equity Options: 9:30 a.m. to 4:00 p.m. (ends at 1:00 p.m. on early closing days).

Trading Session for ETF and Index Options: 9:30 a.m. to 4:15 p.m. (ends at 1:15 p.m. on early closing days).

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1.1.2 Obtaining more information

Information such as (but not limited to) membership, rules, data feeds, fees and support can be obtained by sending an email to Trading Operations or by referring to the MIAX website.

1.2 Testing of AIS Subscription

MIAX can provide testing assistance on MIAX testing area for the AIS Feed and the AIS retransmission interface.

Please contact MIAX Trading Operations at TradingOperations@MIAXOptions.com or (609) 897-7302 to obtain more information about the aforementioned.

1.3 Answers to FAQs

Subscription: Please contact Trading Operations for details about subscribing to AIS.

Symbol and Strategy management: Subscribers to the data feed will get a list of all option symbols and strategies that will be traded and sourced on that feed at the start of every session. If firms cannot start listening to the feed in time for the normal symbol and complex strategy broadcast, they can connect to the AIS Retransmission service and request for a Last Value Refresh Service (see section 0) or request all messages published and then subsequently process only the symbol and strategy messages to build their symbol and/or strategy list. The MIAX assigned Product ID or Strategy ID of each option/strategy will be sent in every message so that firms can tie each message to an option symbol or strategy.

Retransmission: Gap-fill packets generated as a response to retransmission requests are only disseminated on the retransmission TCP channels and not on the real-time multicast feeds.

Redundant Feeds: In order to achieve higher availability, MIAX offers the real-time AIS feed in two separate redundant and identical feeds named “A Feed” and “B Feed”. Firms are advised to arbitrate between the two feeds in order to mitigate gaps and achieve higher availability. “A Feed” is the primary feed from the primary data center and “B Feed” is the secondary feed from the secondary data center.

Refresh Service: Refresh service is provided only on the retransmission TCP channels and does not affect the real-time AIS feed.

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1.4 Data Types

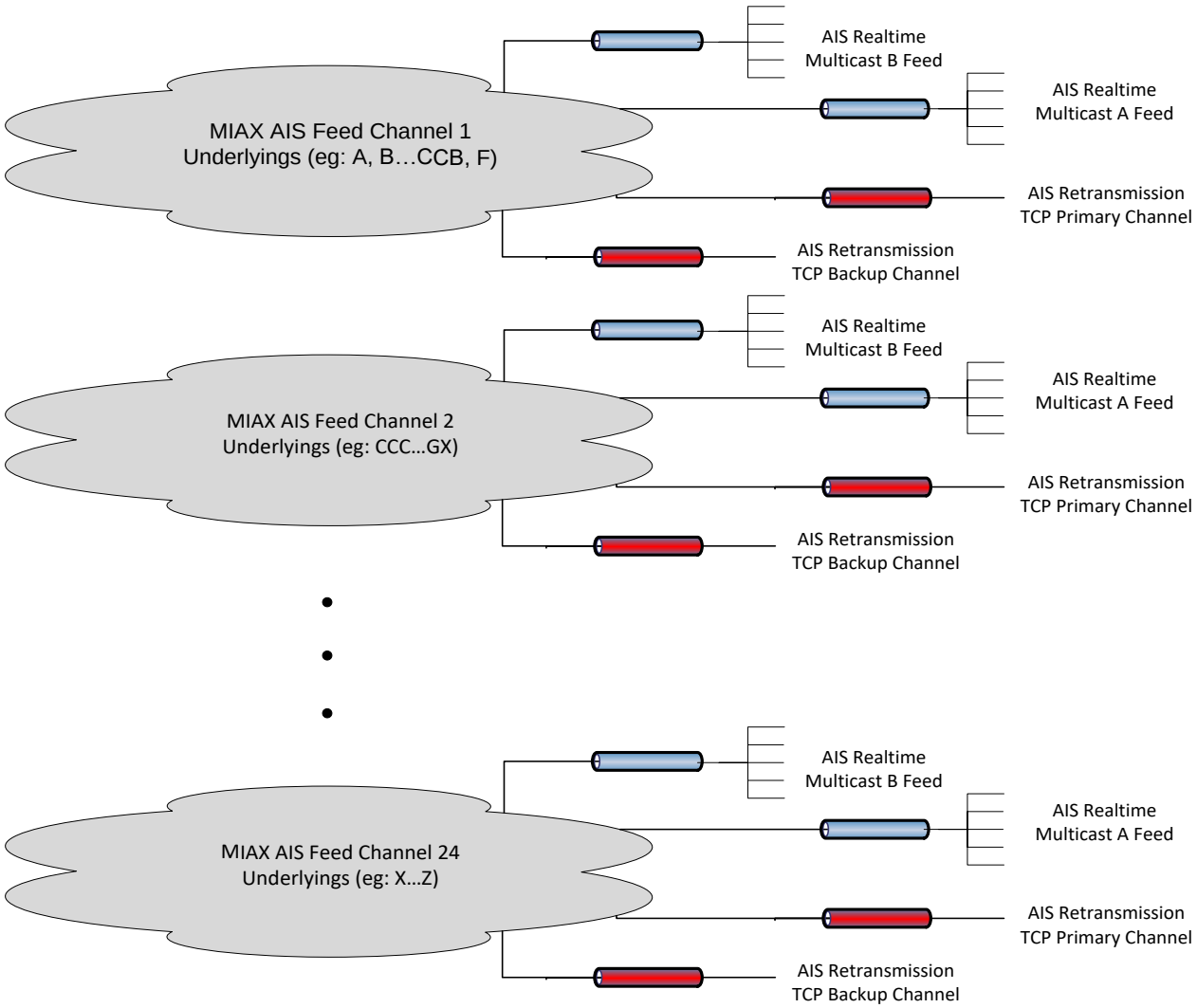
The following table describes the data types used in AIS messaging:
Note: Time fields in all messages are as per timings of United States Eastern Time zone unless specified otherwise.

Data Type	Description
BinaryS	Signed, Intel x86 byte-ordered (**little-endian**), binary encoded numbers
BinaryU	Unsigned, Intel x86 byte-ordered (**little-endian**), binary encoded numbers
BinaryPrc4S	BinaryS Field with the last 4 (right most) digit places being decimal places
BinaryPrc4U	BinaryU Field with the last 4 (right most) digit places being decimal places
BinaryPrc2U	BinaryU Field with the last 2 (right most) digit places being decimal places
SecTime	BinaryU field that contain transaction time in seconds since Epoch (January 1, 1970, 00:00:00 **UTC**)
NanoTime	BinaryU field that contain transaction time in nanoseconds since past second
Alphanumeric	Each place can contain characters or numbers. Left justified and space-padded on the right

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2. AIS Architecture

MIAX Administrative Information Subscriber (AIS) Architecture



Highlights:

- Real-time dissemination is separated out on to 24 separate Feed channels.
- A Feed channel will contain sourced data for all options for a single underlying.
- Any options for any given underlying will only be sourced by a single feed channel on any given day.
- Each Feed channel sources independently from the other groups and hence has independent sequence numbers.
- All the messages on each feed channel will be published in FIFO sequence.

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- High availability is achieved by disseminating identical data on an “A Feed” and “B Feed” for each Feed channel.
- Underlyings may not be contiguously distributed according to symbol ranges in each Feed channel.
- Two separate TCP based retransmission channels for each Feed channel supply AIS retransmission via the AIS Retransmission Interface.

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3. Session Level Protocol

3.1 Real-time AIS Feed

AIS real-time feed uses MIAX's proprietary **MACH protocol**. Each AIS Packet may have multiple application messages and each application message is encapsulated in a MACH protocol packet. Hence a single AIS packet may contain 1 or more sequenced MACH protocol packets.

Please refer to MACH document (available at the MIAX website) for details about MACH protocol. This protocol layer offers low latency application messaging over multicast, sequencing of messages and heartbeats.

3.2 AIS Retransmission Interface

AIS Retransmission Interface uses MIAX's proprietary **SesM – TCP Session Management Protocol**.
Please refer to the latest SesM TCP Session Management document (available at the MIAX website) for details about SesM session management protocol. This protocol layer offers session management capabilities such as authentication, application messaging over TCP/IP, sequencing of messages, heartbeats and gap fills.

Firms must first use the `Login Request` with a requested sequence number of **zero** to login to the Interface. After receiving a successful `Login Response`, the firm can choose either the 3.2.1 SesM Gap Fill Service or 3.2.2 Last Value Refresh Service.

3.2.1 SesM Gap Fill Service

Firms can use the **Retransmission Request** session management message, available in the SesM protocol, to request retransmission of a specific range of packets, identified by sequence numbers.

3.2.2 Last Value Refresh Service

3.2.2.1 Request Message to MIAX

Firms can use the **Unsequenced Data Packet,** available in the SesM protocol, to request a last value refresh of various market data and status information. The Refresh Request has the following format:

Field Name	Length	Data Type	Notes
SesM Packet Length	2	Binary	
SesM Packet Type	1	Alphanumeric	*'U' – SesM Unsequenced Packet*
Request Type	1	Alphanumeric	"R" – Refresh

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Refresh Message Type	1	Alphanumeric	"P" - Simple Series Update Refresh "C" – Complex Strategy Definition Refresh "U" – Underlying Trading Status Refresh "S" – System State Refresh

3.2.2.2 Response Message from MIAX

The Retransmission feed will respond to the Refresh request with a series of SesM-TCP **Unsequenced Data Packets** based on the `Refresh Message Type`. Each response message will have the *following format:*

Field Name	Length	Data Type	Notes
SesM Packet Length	2	Binary	
SesM Packet Type	1	Alphanumeric	*'U' – SesM Unsequenced Packet*
Response Type	1	Alphanumeric	"R" – AIS Refresh
Sequence Number	8	BinaryU	Original sequence number from live feed.
Application Message	varies	See section 0	Based on the message type requested.

The first SesM packet to be received by the firms will be the 4.1 MIAX System Time Message (See section 0). The timestamp (combined with the nanosecond part in the subsequent messages) represents the most recent Matching Engine transaction time. It is ***not*** the original timestamp from the MACH sequenced messages in the live feed. *The sequence number in the refresh messages may be used to arbitrate with the sequenced packets from live feed (eg: data with higher sequence number from either the refresh or the live feed represents latest information).*

3.2.2.3 End of Refresh Notification from MIAX

When the refresh is complete MIAX will send the following message.

Field Name	Length	Data Type	Notes
SesM Packet Length	2	Binary	
SesM Packet Type	1	Alphanumeric	*'U' – SesM Unsequenced Packet*
Response Type	1	Alphanumeric	"E" – End of Request.
Refresh Message Type	1	Alphanumeric	from Refresh Request

3.2.3 Session Termination

After satisfying the retransmission request, AIS Retransmission Interface will send a `Goodbye Packet` and disconnect the TCP connection.
Note: Upon receipt of an unknown, malformed or illegal session message, MIAX will send a SesM "Goodbye Packet" with a human readable reason text string and MIAX will disconnect the line.

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4. Application Message Formats

This section consists of format of messages sent over the AIS feed.

The time specified in the *Timestamp* field in all the messages below is the time at which the Matching Engine associated with that underlying group published the message. This is the same timestamp that will get included in the messages transmitted on the retransmission interface.

4.1 MIAX System Time Message

This is the message format that will be used to disseminate the "seconds" part of the timestamp that is applicable to all messages that are sent in the current second.

Field Name	Length	Data Type	Notes
MACH Protocol Data			*Refer to MACH Protocol Specification*
Message Type	1	Alphanumeric	"1"
Time Stamp	4	SecTime	Seconds part of the time that applies to all messages that gets disseminated until this message gets sent again.

Points to note:

- Note that this message is only sent when there are any application messages that are going to be sent during any second. Firms are advised to not assume that there will be a message for every second of the day.

4.2 Simple Series Update

This is the message format that will be used to disseminate all Option series traded on MIAX for the current session.

Field Name	Length	Data Type	Notes
MACH Protocol Data			*Refer to MACH Protocol Specification*
Message Type	1	Alphanumeric	"P"
Product Add/Update Time	4	NanoTime	Time at which this product is added/updated on MIAX system today.
Product ID	4	BinaryU	MIAX Product ID mapped to a given option. It is assigned per trading session and is valid for that session.
Underlying Symbol	11	Alphanumeric	Stock Symbol for the option.
Security Symbol	6	Alphanumeric	Option Security Symbol
Expiration Date	8	Alphanumeric	Expiration date of the option in YYYYMMDD format
Strike Price	4	BinaryPrc4U	Explicit strike price of the option. Refer to data types for field processing notes

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<table>
<tr><th>Field Name</th><th>Length</th><th>Data Type</th><th>Notes</th></tr>
<tr><td>Call or Put</td><td>1</td><td>Alphanumeric</td><td>Option Type
“C” = Call
“P” = Put</td></tr>
<tr><td>Opening Time</td><td>8</td><td>Alphanumeric</td><td>Expressed in HH:MM:SS format. Eg: 09:30:00</td></tr>
<tr><td>Closing Time</td><td>8</td><td>Alphanumeric</td><td>Expressed in HH:MM:SS format. Eg: 16:15:00</td></tr>
<tr><td>Restricted Option</td><td>1</td><td>Alphanumeric</td><td>“Y” = MIAX will accept position closing orders only
“N” = MIAX will accept open and close positions</td></tr>
<tr><td>Long Term Option</td><td>1</td><td>Alphanumeric</td><td>“Y” = Far month expiration (as defined by MIAX rules)
“N” = Near month expiration (as defined by MIAX rules)</td></tr>
<tr><td>Active on MIAX</td><td>1</td><td>Alphanumeric</td><td>Indicates if this symbol is tradable on MIAX in the current session:
“A” = Active (tradable) on MIAX
“I” = Inactive (not tradable) on MIAX</td></tr>
<tr><td>MIAX BBO Posting Increment Indicator</td><td>1</td><td>Alphanumeric</td><td>This is the Minimum Price Variation as agreed to by the Options industry (penny pilot program) and as published by MIAX
<table>
<tr><th rowspan="2">Indicator</th><th colspan="2">BBO Increments</th></tr>
<tr><th>Price <= $3</th><th>Price > $3</th></tr>
<tr><td>“P”</td><td>Penny (0.01)</td><td>Penny (0.01)</td></tr>
<tr><td>“N”</td><td>Penny (0.01)</td><td>Nickel (0.05)</td></tr>
<tr><td>“D”</td><td>Nickel (0.05)</td><td>Dime (0.10)</td></tr>
</table></td></tr>
<tr><td>Liquidity Acceptance Increment Indicator</td><td>1</td><td>Alphanumeric</td><td>This is the Minimum Price Variation for Quote/Order acceptance as per MIAX rules
<table>
<tr><th rowspan="2">Indicator</th><th colspan="2">Quoting Increments</th></tr>
<tr><th>Price <= $3</th><th>Price > $3</th></tr>
<tr><td>“P”</td><td>Penny (0.01)</td><td>Penny (0.01)</td></tr>
<tr><td>“N”</td><td>Penny (0.01)</td><td>Nickel (0.05)</td></tr>
<tr><td>“D”</td><td>Nickel (0.05)</td><td>Dime (0.10)</td></tr>
</table></td></tr>
</table>

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<table>
<tr><th>Field Name</th><th>Length</th><th>Data Type</th><th>Notes</th></tr>
<tr><td>Opening Underlying Market Code</td><td>1</td><td>Alphanumeric</td><td>Options opening will be triggered on receipt of Opening quote/trade from this Underlying market:
<table>
<tr><th>Market Code</th><th>Description</th></tr>
<tr><td>A</td><td>NYSE Amex</td></tr>
<tr><td>B</td><td>NASDAQ OMX BX</td></tr>
<tr><td>C</td><td>National Stock Exchange</td></tr>
<tr><td>D</td><td>FINRA ADF</td></tr>
<tr><td>E</td><td>Market Independent (Any market that opens first)</td></tr>
<tr><td>F</td><td>TXSE</td></tr>
<tr><td>H</td><td>MIAX PEARL Equities</td></tr>
<tr><td>I</td><td>International Securities Exchange</td></tr>
<tr><td>J</td><td>EDGA Exchange, Inc</td></tr>
<tr><td>K</td><td>EDGX Exchange, Inc</td></tr>
<tr><td>L</td><td>LTSE</td></tr>
<tr><td>M</td><td>Chicago Stock Exchange</td></tr>
<tr><td>N</td><td>NYSE Euronext</td></tr>
<tr><td>P</td><td>NYSE Arca Exchange</td></tr>
<tr><td>Q</td><td>NASDAQ OMX (via UTP Feed)</td></tr>
<tr><td>T</td><td>NASDAQ OMX (via CTA Feed)</td></tr>
<tr><td>U</td><td>MEMX</td></tr>
<tr><td>V</td><td>IEX</td></tr>
<tr><td>W</td><td>CBOE Stock Exchange (CBSX)</td></tr>
<tr><td>X</td><td>NASDAQ OMX PHLX</td></tr>
<tr><td>Y</td><td>BATS Y-Exchange, Inc</td></tr>
<tr><td>Z</td><td>BATS Exchange Inc</td></tr>
</table></td></tr>
<tr><td>Priority Quote Width</td><td>4</td><td>BinaryPrc4U</td><td>Maximum allowable width for a quote for this Option during regular trading in order to be considered as Priority Quote. 0 when Priority Quote Width is not applicable.</td></tr>
<tr><td>Reserved</td><td>8</td><td>BinaryU</td><td>** Reserved for future use **</td></tr>
</table>

Points to note:

- Entire Options list will be disseminated at the start of day.
- In each channel, firms will only receive the series associated with the Engine that is servicing that channel.
- Intra-day updates will also be published as they occur.
- In case of an intra-day reconnection, users can request all Options series data from the AIS retransmission line.
- The Priority quote width specified in this message is not applicable to Opening. Please refer MIAX rules and circulars for details about priority quote width applicable during Opening.

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4.3 Complex Strategy Definition Update

This is the message format that will be used to disseminate stock option strategies traded on MIAX for the current session. The Strategy ID sent in this message is utilized by the Administrative Information Subscriber (AIS) Feed for Complex Liquidity Seeking Events, the MIAX Order Feed (MOR) for Complex Order dissemination, the Complex Top of Market Feed (cToM) for Complex Trade and Top of Market dissemination and the MIAX Express Interface (MEI) for Complex eQuotes.

Field Name	Length	Data Type	Notes
MACH Protocol Data			*Refer to MACH Protocol Specification*
Message Type	1	Alphanumeric	"C"
Strategy Add Time	4	NanoTime	Time at which this Strategy is added/updated on MIAX system today.
Strategy ID	4	BinaryU	MIAX Strategy ID is assigned per trading day and is valid only for that day.
Underlying Symbol	11	Alphanumeric	Underlying Symbol for this strategy
Active on MIAX	1	Alphanumeric	Indicates if this strategy is tradable on MIAX in the current session: "A" = Active (tradable) on MIAX "I" = Inactive (not tradable) on MIAX
Reserved	1	BinaryU	** Reserved for future use **
Update Reason	1	Alphanumeric	"N" – New strategy created "U" – Updated
Reserved	10	BinaryU	** Reserved for future use **
Number of Legs	1	BinaryU	Number of Legs. Variable from 2 to 8
➡ **Product ID**	4	BinaryU	*Option leg:* MIAX Defined Series. See Simple Series Update Message. *Stock leg:* 0 (zero)
➡ **Leg Ratio Qty**	2	BinaryU	The ratio of this individual leg. Number of option contracts or Number of stock shares for this leg is: LegRatioQty * OrderQty

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Field Name	Length	Data Type	Notes
➡ **Leg Side**	1	Alphanumeric	The side of this individual leg Valid values are: “B” = Bid “A” = Ask
➡ **Reserved**	8	BinaryU	** Reserved for future use **

Points to note:

- Strategies may be created intra-day as orders are placed at the MIAX Exchange or pre-defined before the market open.
- In each channel, firms will only receive the Strategies associated with the Engine that is servicing that channel.
- In case of an intra-day reconnection, users can request all Strategies definitions from the AIS retransmission lines.
- The length of this message is ***variable*** based on the number of legs.
- When underlying halts, all strategies for that underlying are in a halted state. Firms should process Underlying Trading Status notification to determine current state of the strategies.
- This message might be published more than once per day. When Update Reason is “U”, the only field that can change is “Active on MIAX”.
- The tradability of a strategy can be tracked with the status of the underlying (message type “H”) or the individual series. (message type “P”)
- The Strategy ID and Product ID fields are separate and distinct fields with assigned ID’s per trading day and valid only for the current day. Their scope is limited to each field.

4.4 System State

This message format is used to notify the firms of the state changes of the system. This is a notification that applies to each Underlying group. Firms can use notifications as triggers in their system to ensure electronic synchronization of systems.

Field Name	Length	Data Type	Notes
MACH Protocol Data			*Refer to MACH Protocol Specification*
Message Type	1	Alphanumeric	“S”
Notification Time	4	NanoTime	Time at which this was generated by MIAX system.
AIS Version	8	Alphanumeric	Eg: AIS2.0
Session ID	4	BinaryU	MIAX assigned ID for the current trading session

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System Status	1	Alphanumeric	Current system status: “S” = Start of System hours “C” = End of System hours “1” = Start of Test Session (sent before tests). “2” = End of Test Session.

Points to note:

- Firms must ensure that messages sent on the AIS Feed from the beginning of “start of test session” to the end of “end of test session” will not affect their production session while allowing the firms to still be involved in production tests and dry runs.
- A change in Session ID will mean that restarting at MACH sequence number 1 for that Underlying group. Refer to MACH protocol specification for details about this. Firms must be able to handle more than one trading session in a single trading day.

4.5 Simple Liquidity Seeking Event Notification

This is the message format that will be used to disseminate information each time a Liquidity seeking event (defined in Event Type field below) starts at MIAX.

Field Name	Length	Data Type	Notes
MACH Protocol Data			*Refer to MACH Protocol Specification*
Message Type	1	Alphanumeric	“L”
Timestamp	4	NanoTime	Time at which this notification was generated by MIAX system.
Product ID	4	BinaryU	MIAX Product ID mapped to a given option. It is assigned per trading session and is valid for that session.
Event Type	1	Alphanumeric	Type of Event: “L” – Liquidity Refresh Mechanism “O” – Opening/Reopening Imbalance Mechanism “R” – Route Mechanism “P” – MIAX PRIME Paired order “S” – Settlement Opening Imbalance Mechanism “E” – Liquidity Exposure Process
Event ID	4	BinaryU	Values: When Event Type is ‘O’: always 0 When Event Type is ‘L’, ‘R’, or ‘P’: Unique Event ID for the current event.
Price	4	BinaryPrc4U	Current price of the Event. When EventType is ‘P’: Auction Start Price not disseminated (will always be 0) Usage: Please refer to Appendix C for details.

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Field Name	Length	Data Type	Notes
Imbalance Side	1	Alphanumeric	This is the side of the excess liquidity that is waiting to get matched 'B' = Bid 'A' = Ask Usage: When Event Type is 'O': Imbalance Side. When Event Type is 'R': Side that will be routed. When Event Type is 'L': Side of oversized order initiating liquidity refresh. When Event Type is 'P': Agency side of MIAX PRIME order being auctioned. Please refer to Appendix C for details.
Quantity 1	4	BinaryU	Usage: When Event Type is 'O', 'L', 'R', or 'P': MatchQty – Number of contracts ready to be matched (both buyers and sellers already present). Other Event Types: 0 (zero) Please refer to Appendix C for details.
Quantity 2	4	BinaryU	Usage: When Event Type is 'O', 'L' or 'R': Routable Quantity. Number of contracts that may be routed. Other Event Types: 0 (zero) Please refer to Appendix C for details.
Quantity 3	4	BinaryU	Usage: When Event Type is 'O','L' or 'R': Imbalance Quantity. Number of contracts on the *Imbalance Side* that cannot be matched. Other Event Types: 0 (zero) Please refer to Appendix C for details.
Quantity 4	4	BinaryU	Usage: When Event Type is 'O', 'L' or 'R': Must Fill Quantity. Number of contracts, on the provided *Side*, that must be matched at the provided *Price*. These contracts are crossing the provided *Price* and MIAX is seeking liquidity for these contracts. Other Event Types: 0 (zero) Please refer to Appendix C for details.
Attributable ID	4	Alphanumeric	MIAX assigned Attributable ID (Executing Broker MPID) of an Order. Otherwise, it is space filled.

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Field Name	Length	Data Type	Notes
Origin	1	Alphanumeric	Origin of either the single liquidity which initiated the event or origins of multiple liquidities which initiated the event if they are all of identical origin. "1" - Market Maker "2" - Away Market Maker "3" - Broker Dealer "4" - Firm "5" - Priority Customer "6" – Non-Priority Customer " " - multiple initiators of varying origin
Reserved	7	BinaryU	** Reserved for future use. **

Points to note:

- If there are any changes to the imbalance and if the functionality calls for an update to be published, firms will be notified using the same message format. Please refer to Appendix C for details.
- Firms must use the Event ID published in this notification when responding with Orders or Quotes.

4.6 Complex Liquidity Seeking Event Notification

This is the message format that will be used to disseminate information each time a Liquidity seeking event (defined in Event Type field below) starts at MIAX

Field Name	Length	Data Type	Notes
MACH Protocol Data			*Refer to MACH Protocol Specification*
Message Type	1	Alphanumeric	"l" (lower case L)
Timestamp	4	NanoTime	Time at which this notification was generated by MIAX system.
Strategy ID	4	BinaryU	MIAX Strategy ID is assigned per trading day and is valid only for that day.
Event Type	1	Alphanumeric	Type of Event: "C" – Complex Order Auction "P" – cPRIME Auction "E" – Complex Liquidity Exposure Process Auction
Event ID	4	BinaryU	Unique Event ID for the current event.
Side	1	Alphanumeric	"B" – Bid "A" – Ask When Event Type is 'P': Agency side of the cPRIME order being auctioned

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Field Name	Length	Data Type	Notes
Price	8	BinaryPrc4S	The net limit price for the strategy If Side is “B”: • Positive number represents net debit • Negative number represents net credit If Side is “A”: • Positive number represents net credit • Negative number represents net debit Price of zero is net neutral transaction for either side. When Event Type is ‘P’: Auction Start Price not disseminated (will always be 0) Usage: Please refer to Appendix C for details.
Matched Quantity	4	BinaryU	Usage: Number of strategies ready to be matched (both buyers and sellers already present).
Imbalance Quantity	4	BinaryS	Usage: When Event Type is ‘C’: Number of strategies on the *Imbalance Side* that cannot be matched. Other Even Types: 0 (zero)
Attributable ID	4	Alphanumeric	MIAX assigned Attributable ID (Executing Broker MPID) of the Order. Otherwise, it is space filled.
Origin	1	Alphanumeric	Origin of either the single liquidity which initiated the event or origins of multiple liquidities which initiated the event if they are all of identical origin. “1” - Market Maker “2” - Away Market Maker “3” - Broker Dealer “4” - Firm “5” - Priority Customer “6” – Non-Priority Customer “ “ - multiple initiators of varying origin
Reserved	7	BinaryU	** Reserved for future use. **

Points to note:

- Firms must use the Event ID published in this notification when responding with auction responses (Orders or eQuotes).

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4.7 Underlying Trading Status Notification

This message format will be used to notify firms of changes to the trading status of all the options of an underlying.

Message Direction: MIAX to Firm

Field Name	Length	Data Type	Notes
MACH Protocol Data			*Refer to MACH Protocol Specification*
Message Type	1	Alphanumeric	"H"
Timestamp	4	NanoTime	Time at which this was generated by MIAX system.
Underlying Symbol	11	Alphanumeric	Underlying Symbol
Trading Status	1	Alphanumeric	"H" = MIAX has halted trading for this Underlying Symbol "R" = MIAX will resume trading (reopen) for this Underlying Symbol "O" = MIAX will open trading for this Underlying Symbol "C" = MIAX has closed trading for this Underlying Symbol
Event Reason	1	Alphanumeric	"A" = This event resulted from automatic/market driven event "M" = MIAX manually initiated this event
Expected Event Time: Seconds Part	4	SecTime	Seconds portion of the expected time of the event. Always use in conjunction with the Nano-seconds part field.
Expected Event Time: Nano-Seconds Part	4	BinaryU	Nano-seconds portion of the expected time of the event. Specifies number of nano-seconds since the seconds specified in "Expected Event Time Seconds" field.

Points to note:

- When underlying trading status ="H" or "C", Expected Event Time Seconds/Nano-Seconds will be set to 0 (zero).
- When underlying trading status = "R" or "O", Expected Event Time (Seconds/Nano-Seconds Parts) will be set to the time at which the opening/reopening process will start for this Underlying Symbol.

4.8 Theoretical Settlement Reference Price Notification

This message format will be used to notify firms of the current state and Theoretical Settlement Reference Price (TSRP) or Final Settlement Reference Price (FSRP) of each option involved in settlement, on the settlement day, until the Final Settlement Price is determined.

Message Direction: MIAX to Firm

Field Name	Length	Data Type	Notes
MACH Protocol Data			*Refer to MACH Protocol Specification*
Message Type	1	Alphanumeric	"M"
Timestamp	4	NanoTime	Time at which this was generated by MIAX system.

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Field Name	Length	Data Type	Notes
Product ID	4	BinaryU	MIAX Product ID mapped to a given option. It is assigned per trading session and is valid for that session.
Underlying Symbol	11	Alphanumeric	Stock Symbol for the option
Security Symbol	6	Alphanumeric	Option Security Symbol
Expiration Date	8	Alphanumeric	Expiration date of the option in YYYYMMDD format
Strike Price	4	BinaryPrc4U	Strike Price of the Option
Call or Put	1	Alphanumeric	Option Type 'C' – Call 'P' – Put
Option State	1	Alphanumeric	Option State 'P' – Pre-Open 'N' – Open, but no FSRP 'O' – Open with FSRP
Prior Day Reference Price	4	BinaryPrc4U	Reference Price of the option used for Index calculation at the end of the prior trading day
Settlement Reference Price	4	BinaryPrc4U	Current Settlement Reference Price (SRP) of the Option could be theoretical (TSRP) or final (FSRP)
Settlement Reference Price Type	1	Alphanumeric	Type of the Settlement Reference Price: 'E' – Expected Opening Price [TSRP] 'A' – Auction Only Price [TSRP] 'O' – Opening Price [TSRP] 'S' – Final Settlement Reference Price [FSRP]
SAO Buy Quantity 1	4	BinaryU	All open SAO Buy liquidity priced equal to or better than the Theoretical Settlement Reference Price, including Market Orders
SAO Sell Quantity 1	4	BinaryU	All open SAO Sell liquidity priced equal to or better than the Theoretical Settlement Reference Price, including Market Orders
SAO Buy Quantity 2	4	BinaryU	All open SAO Buy liquidity at all prices
SAO Sell Quantity 2	4	BinaryU	All open SAO Sell liquidity at all prices
Non SAO Buy Quantity	4	BinaryU	All open non-SAO Buy liquidity priced equal to or better than the Theoretical Settlement Reference Price, including Market Orders
Non SAO Sell Quantity	4	BinaryU	All open non-SAO Sell liquidity priced equal to or better than the Theoretical Settlement Reference Price, including Market Orders
Total Buy Quantity	4	BinaryU	Total SAO and non-SAO open Buy liquidity priced equal to or better than the Theoretical Settlement Reference Price, including Market Orders

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Field Name	Length	Data Type	Notes
Total Sell Quantity	4	BinaryU	Total SAO and non-SAO open Sell liquidity priced equal to or better than the Theoretical Settlement Reference Price, including Market Orders
Imbalance Side	1	Alphanumeric	This is the side of the excess liquidity that is waiting to get matched ‘ ‘ (space) = No Imbalance ‘B’ = Bid ‘A’ = Ask
Imbalance Quantity	4	BinaryU	Number of contracts on the Imbalance Side that cannot be matched either at a price equal or better than the Theoretical Settlement Reference Price or at expected opening price.
Must Fill Quantity	4	BinaryU	Number of unmatched contracts, on the provided Side, that must be matched at the price better than the Theoretical Settlement Reference Price.
Matched Quantity	4	BinaryU	Number of contracts ready to be matched (both buyers and sellers already present) at the Theoretical Settlement Reference Price.
Opening Condition	1	Alphanumeric	Current Opening Condition: ‘Q’ – Need Quote ‘C’ – Crossed Quote ‘A’ – Crossed Away Quote ‘B’ – Need Buyers (when Imbalance Side = ‘A’) ‘S’ – Need Sellers (when Imbalance Side = ‘B’) ‘O’ – Would Open ‘N’ – Open with no Final SRP ‘R’ – Open with Final SRP
MBB	4	BinaryPrc4U	MIAX Best Quote Bid
MBO	4	BinaryPrc4U	MIAX Best Quote Offer
Reserved	8	BinaryU	** Reserved for future use. **

Points to note:

- The Theoretical Settlement Reference Price message will be sent via AIS at a regular interval (e.g. 60 secs or 10 secs) until the Final Settlement Price is determined.
- Before an option opens, the Settlement Reference Price will always be theoretical for that option. After the open it can be theoretical or final, this will be indicated by the Settlement Price Type field.

4.9 Theoretical Settlement Price Notification

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This message format will be used to notify firms of the Theoretical Settlement Price, on the settlement day, until the Final Settlement Price is determined.

Message Direction: MIAX to Firm

Field Name	Length	Data Type	Notes
MACH Protocol Data			*Refer to MACH Protocol Specification*
Message Type	1	Alphanumeric	"N"
Timestamp	4	NanoTime	Time at which this was generated by MIAX system.
Settlement Symbol	8	Alphanumeric	Settlement Symbol
Theoretical Settlement Price	4	BinaryPrc2U	Theoretical Settlement Price
Reserved	16	BinaryU	** Reserved for future use **

Points to note:

- The Theoretical Settlement Price is calculated at a regular interval (e.g. 60 secs or 10 secs) until the final Settlement Price is determined.
- Final Settlement Price is only sent on MPF. Theoretical Settlement Price is only sent on AIS

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Appendix A: MIAX AIS Subscription/Connectivity Information

Please visit the MIAX website to obtain the most up-to-date information about the following:

- Real-time Feed multicast groups, ports for A Feed and B Feed.
- Retransmission IP addresses and ports for primary and backup channels.

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Appendix B: Contact List

Please visit the MIAX website to obtain the most up-to-date contact list and other such information.

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Appendix C: Liquidity Seeking Event Notification Details

This section describes the content of price, side and quantity fields in the Liquidity Event Notification Message generated by the system.

Simple Liquidity Event Notification

Prior to 9:30 or Quoting Qualification for Opening (Event Type 'O')	Price	Imbalance Side	Matched Quantity (Message Quantity #1)	Route Quantity (Message Quantity #2)	Imbalance Quantity (Message Quantity #3)	Must Fill Quantity (Message Quantity #4)
Imbalance locking or priced through the Quote Range, but not priced through the Anticipated Opening Price on MIAX	0	Side that cannot be filled locking or priced through Quote Range	Number of contracts matched on MIAX at Price	0	Number of contracts that cannot be filled by opposing liquidity when locking or priced through Price	0
Imbalance priced through the Anticipated Opening Price on MIAX	0	Side priced through Expanded Quote Range	Number of contracts matched on MIAX at Price	0	Number of contracts that cannot be filled by opposing liquidity when locking or priced through Price	Number of contracts that cannot be filled by opposing liquidity when priced through the EQR price

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Imbalance comprised of Market Order(s) with no valid width quotes present (No valid Quote Range or Expanded Quote Range)	0	Side of Larger Market Order	Number of contracts matched on MIAX by opposing market order(s)	0	Number of contracts that cannot be filled by opposing liquidity	Number of contracts that cannot be filled by opposing market order(s)

After 9:30 with Quoting Qualification for Opening (Event Type 'R')	**Price**	**Imbalance Side**	**Matched Quantity (Message Quantity #1)**	**Route Quantity (Message Quantity #2)**	**Imbalance Quantity (Message Quantity #3)**	**Must Fill Quantity (Message Quantity #4)**
Opening Route	Best price to which opening interest will be routed	Side priced through Away BBO(s)	Number of contracts matched on MIAX at Price	Number of contracts MIAX will route at Price	Number of contracts that cannot be filled by opposing liquidity when locking or priced through Price	Number of contracts that cannot be filled by opposing liquidity when priced through the EQR price

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After 9:30 with Quoting Qualification for Opening (Event Type 'O')	Price	Imbalance Side	Matched Quantity (Message Quantity #1)	Route Quantity (Message Quantity #2)	Imbalance Quantity (Message Quantity #3)	Must Fill Quantity (Message Quantity #4)
Opening Imbalance Crossing Quote Range or non-Quality Market, not priced through Anticipated Opening Price or Expanded Quote Range (OpenPrice>QR & <EQR or Open Width>Quality)	Anticipated Opening price	Side priced through Quote Range	Number of contracts matched on MIAX at Price	Number of contracts MIAX will route at Price	Number of contracts that cannot be filled by opposing liquidity when locking or priced through Price	0
Opening Imbalance locking or Crossing ABBO, but not Crossing MIAX Quote Range or Expanded Quote Range	Best price of unmatched liquidity	Side of unmatched liquidity	Number of contracts matched on MIAX at Price	Number of contracts MIAX will route at Price or better	Unmatched liquidity at Price	0
Opening Imbalance Locking or Crossing Anticipated Opening Price or Expanded Quote Range	Anticipated Opening price	Side locking or priced through the Anticipated Opening Price	Number of contracts matched on MIAX at Price	Number of contracts MIAX will route at Price	Number of contracts that cannot be filled by opposing liquidity when locking or priced through Price	Number of contracts that cannot be filled by opposing liquidity when priced through the EQR Price

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MIAX Quotes Cross Quotes, do not Cross Orders or ABBO	Anticipated Opening price	Side of Larger Quote	Number of contracts matched on MIAX at Price	0	0	0

After 9:30 with Quoting Qualification for Opening (Event Type 'S')	Price	Imbalance Side	Matched Quantity (Message Quantity #1)	Route Quantity (Message Quantity #2)	Imbalance Quantity (Message Quantity #3)	Must Fill Quantity (Message Quantity #4)
Special Settlement Opening Imbalance Locking or Crossing Anticipated Opening Price or Expanded Quote Range	Anticipated Opening price	Side locking or priced through the Anticipated Opening Price	Number of contracts matched on MIAX at Price	Number of contracts MIAX will route at Price	Number of contracts that cannot be filled by opposing liquidity when locking or priced through Price	Number of contracts that cannot be filled by opposing liquidity when priced through the EQR price

Free Trading – Route (Event Type 'R')	Price	Imbalance Side	Matched Quantity (Message Quantity #1)	Route Quantity (Message Quantity #2)	Imbalance Quantity (Message Quantity #3)	Must Fill Quantity (Message Quantity #4)
Route (Published once per event, no updates)	Transaction Reference Price (TRP)	Initiating Order Side	Number of contracts matched on the MIAX at the next available MIAX price	Number of contracts MIAX will route at the TRP	Number of contracts that cannot be filled and are locking or priced through the TRP	Number of contracts that cannot be filled and are priced through the TRP

Free Trading – Liquidity Refresh (Event Type 'L')	Price	Imbalance Side	Matched Quantity (Message Quantity #1)	Route Quantity (Message Quantity #2)	Imbalance Quantity (Message Quantity #3)	Must Fill Quantity (Message Quantity #4)
Liquidity Refresh (Published once per event, no updates)	Transaction Reference Price (TRP)	Initiating Order Side	Number of contracts matched on the MIAX the next available MIAX price	0	Number of contracts that cannot be filled and are locking or priced through the TRP	Number of contracts that cannot be filled and are priced through the TRP

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Free Trading – Liquidity Refresh (Event Type 'E')	Price	Imbalance Side	Matched Quantity (Message Quantity #1)	Route Quantity (Message Quantity #2)	Imbalance Quantity (Message Quantity #3)	Must Fill Quantity (Message Quantity #4)
Liquidity Exposure Process Timer (LEPT) (Published once per event, no updates)	Price at which the order is resting on the book during the Timer	Initiating Order Side	Number of contracts matched on the MIAX at the next available MIAX price	0	Number of contracts that cannot be filled and are locking or priced through the Price	Number of contracts that cannot be filled and are through the Price

Free Trading – MIAX PRIME Auction (Event Type 'P')	Price	Imbalance Side	Matched Quantity (Message Quantity #1)	Route Quantity (Message Quantity #2)	Imbalance Quantity (Message Quantity #3)	Must Fill Quantity (Message Quantity #4)
MIAX PRIME Auction (published once per event, no updates)	MIAX PRIME Auction Start Price not disseminated (will always be 0)	MIAX PRIME Agency Order Side	MIAX PRIME Auction Size	0	0	0

Complex Liquidity Event Notification

Free Trading – Complex Strategy Auction (Event Type 'C')	Price	Side	Matched Quantity (Message Quantity #1)	Route Quantity (Message Quantity #2)	Imbalance Quantity (Message Quantity #3)	Must Fill Quantity (Message Quantity #4)
Complex Order Auction (Published once per event, no updates)	The order's limit price, or the opposite side displayed Complex MBBO as derived from the Simple Market if the order is a market order or has a limit price that is equal to, or through, the opposite side displayed Complex MBBO.	The Auction Side is the side that meets the requirements to begin an Auction.	The number of strategies that are matched at the Price without matching against liquidity in the simple order market	NA	The number of strategies that are not matched at the Best Price without matching against liquidity in the Simple Market. Note: A negative Imbalance Quantity indicates an	NA

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					excess of liquidity at the Auction Price on the contra-side.	

Free Trading – Complex Strategy Auction (Event Type 'E')	Price	Side	Matched Quantity (Message Quantity #1)	Route Quantity (Message Quantity #2)	Imbalance Quantity (Message Quantity #3)	Must Fill Quantity (Message Quantity #4)
Complex Liquidity Exposure Process Auctions (cLEPA)	Auction Start Price	Side of the Order that started the cLEPA	The number of strategies that are matched at the Price without matching against liquidity in the simple order market	0	The number of strategies that are not matched at the Best Price without matching against liquidity in the Simple Market.	0

Free Trading – cPRIME Auction (Event Type 'P')	Price	Side	Matched Quantity (Message Quantity #1)	Route Quantity (Message Quantity #2)	Imbalance Quantity (Message Quantity #3)	Must Fill Quantity (Message Quantity #4)
cPRIME Auction (Published once per event, no updates)	cPRIME Auction Start Price not disseminated (will always be 0)	cPRIME Agency Order Side	cPRIME Auction Size	NA	0	NA

Notes:
Transaction Reference Price (TRP): The reference price used during timers and other non-timer transactions.

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Appendix D: Revision History

Revision Date	Version	Description
Nov 03, 2014	1.0	First release.
Jun 29, 2015	1.1	AIS Refresh feature
Jan 15, 2016	1.2	Underlying Market Code: Added 'V' for IEX in Series Update message
Feb 01, 2016	2.0	Added Strategy Update and Strategy Liquidity Seeking Event Notification. Added Complex Strategy Auction to appendix C
Apr 12, 2016	2.0a	Updated field AIS Version in the system state message
Jun 24, 2016	2.1	Update the Free Trading – Complex Strategy Auction (Event Type 'C') in Appendix C
Feb 27, 2017	2.1a	System startup time moved up
Mar 20, 2017	2.2	Added the following in Complex Liquidity Seeking Event Notification message: A new event type (Event Type = "P") A new field "Attributable ID" using 4 bytes of the existing 12 bytes Reserved field at the end of the message Added cPRIME Auction to Appendix C Wrong packet length notes removed from Last value Refresh Service request.
Mar 16, 2018	2.3	Updated Complex Strategy Definition message (Message Type = "C") to support stock-tied strategy definitions
May 08, 2018	2.4	Support for *Settlement Opening Imbalance* and *Liquidity Exposure Process* in *Simple Liquidity Seeking Event Notification* and *Appendix C;* Support for *Complex Liquidity Exposure Process Auction* in *Complex Liquidity Event Notification* and *Appendix C.*
Aug 12, 2020	2.4a	Added new Equities Exchanges
Aug 01, 2022	2.5	Support for Theoretical Settlement Reference Price (Message Type = "M") and Theoretical Settlement Price (Message Type = "N") Notification.
Mar 02, 2026	2.6	Origin field added to both Simple Liquidity Seeking Event Notification and Complex Liquidity Seeking Event Notification
Jul 21, 2026	2.6a	Auction Start Price not disseminated (will always be 0) in Simple Liquidity Seeking Event Notification ("L") and Complex Liquidity Seeking Event Notification ("l" – small L) messages for Event Type "P"
Oct 14, 2026	2.6b	Added Trading Status "C" to Underlying Trading Status Notification message. Added Opening Underlying Market Code "F" (TXSE) to Simple Series Update message

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